1. Problem you were solving

The challenge involves identifying semantic differences between parallel Swiss government documents in different languages. Your goal was to detect differences such as:

- Information added in one language but missing in another.
- Information omitted from one version.
- Changes in meaning between translated documents.

The task uses annotated documents to evaluate how accurately a model identifies these differences at the token level.

2. Approach 1: ModernBERT

You built an encoder-based pipeline using ModernBERT.

Your workflow:

1. Data preparation: Prepared projected German training examples for the encoder.
2. Fine-tuning: Trained ModernBERT-base on the prepared data for four epochs.
3. Prediction: Used the encoder pipeline to generate predictions on validation documents.
4. Evaluation: Evaluated the predictions against annotated validation data.

Your code was organized in the original ~/SwissGov-RSD project:

File	Purpose
encoders/finetuning/data.py	Training-data preparation
encoders/finetuning/train_modernbert.py	Model fine-tuning
encoders/encoder_recognizer.py	Encoder recognition pipeline
scripts/predict_encoder.py	Generating predictions
scripts/evaluate_predictions_admin_de_val.py	Evaluating predictions

Your preliminary evaluation reported Spearman correlation of 0.118 and Kendall's Tau-b of 0.095 on 34 validation inferences. These are preliminary results, not evidence of strong performance.

3. Approach 2: Apertus 8B Instruct

You also explored a generative language-model approach using Apertus 8B Instruct.

Your workflow:

1. Prepared labelled training examples, including augmented examples.
2. Used 4-bit quantization to reduce the model's memory requirements.
3. Fine-tuned the model with QLoRA, using a maximum sequence length of 1,024 tokens.

4. Completed a short training run of 20 steps.
5. Tested the generated predictions using validation scripts.

Your main scripts were:

- scripts/apertus/train_qlora.py
- scripts/apertus/validate_qlora.py

The training run completed, but validation produced repetitive scores and formatting or token-count mismatches. Therefore, you did not establish that fine-tuning improved prediction quality.

4. What you learned from the experiments

The two approaches highlighted different engineering challenges:

- ModernBERT: An encoder-based approach that requires correctly prepared training labels and a reliable prediction and evaluation pipeline.
- Apertus: A generative approach that requires careful prompting, output parsing, token alignment, and validation in addition to fine-tuning.

A model finishing training does not necessarily mean it solves the task successfully. Reliable evaluation and correctly formatted predictions are essential.